Damped Oscillation &

Forced Oscillation

MD. GULAM MUSTAFA

Lecturer (Physics)

Department of Sc. & Hum.

Bangladesh Army International University of Science and Technology (BAIUST)

Syedpur, Cumilla-3501

Whatever we will discuss

- ✓ Types of oscillations
- ✓ Damped oscillation
- ✓ Differential equation of Damped Harmonic Oscillation (DHO)
- ✓ Solution of differential equation of Damped Harmonic Oscillation (DHO)
- ✓ Forced oscillation

Types of oscillations

There are different types of oscillations.

- Free oscillations
- Damped oscillations
- Maintained oscillations
- Forced oscillations

Types of oscillations

Free oscillations A body vibrates with its own natural frequency.

Examples

- Vibrations of tuning fork
- Vibrations in a stretched string
- Oscillations of simple pendulum

Damped oscillations

When the oscillator vibrates in a medium,

- **energy is dissipated in each vibration in overcoming the opposing frictional or viscous forces.**

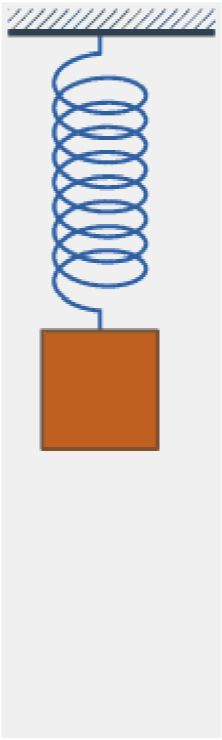
- **the amplitude of vibration, therefore, goes on decreasing progressively with time**

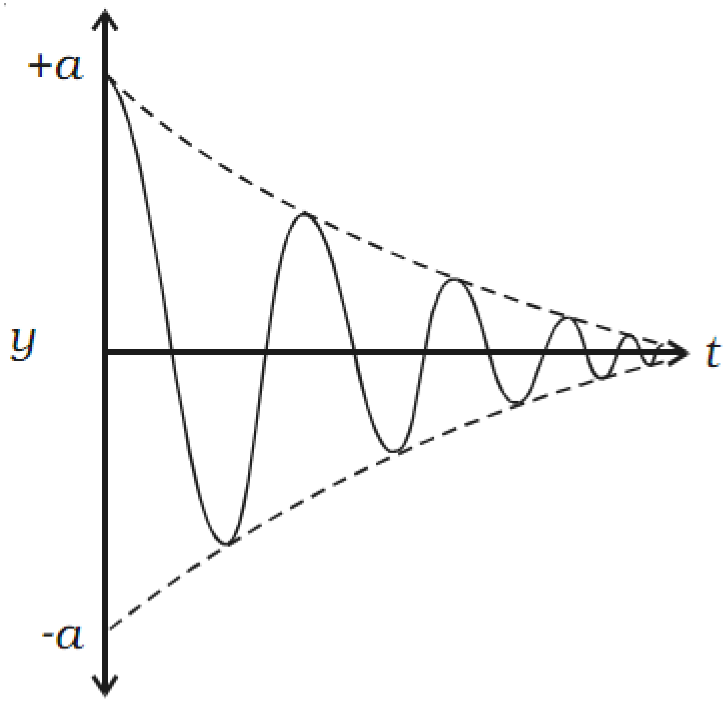
- **finally the oscillations die out**

Such vibrations are called *damped vibrations*.

Damped oscillations

The non-conservative forces have damping effect on the oscillations is damping, resistive or dissipative forces.





Pictorial and graphical representation of Free Damped Vibrations

Damped oscillations

The differential equation of motion of a body executing damped harmonic oscillations can be derived as

$$F = F_r + F_d$$

$$or, m\frac{d^2y}{dt^2} = -ay - bv$$

$$or, m\frac{d^2y}{dt^2} = -ay - b\frac{dy}{dt}$$

$$or, \frac{d^2y}{dt^2} + \frac{b}{m}\frac{dy}{dt} + \frac{a}{m}y = 0$$

$$\boxed{\begin{aligned}\frac{b}{m} &= 2\lambda \\ \frac{a}{m} &= w^2\end{aligned}}$$

Damped oscillations

$$\frac{b}{m} = 2\lambda$$

$$\frac{a}{m} = w^2$$

$$\frac{d^2y}{dt^2} + 2\lambda\frac{dy}{dt} + \omega^2 y = 0$$

Above equation is referred to as the differential equation of a damped harmonic oscillator.

Ref.: Physics for Engineers (Part-1) -Dr. Gias Uddin Ahmed; pp: 81-89

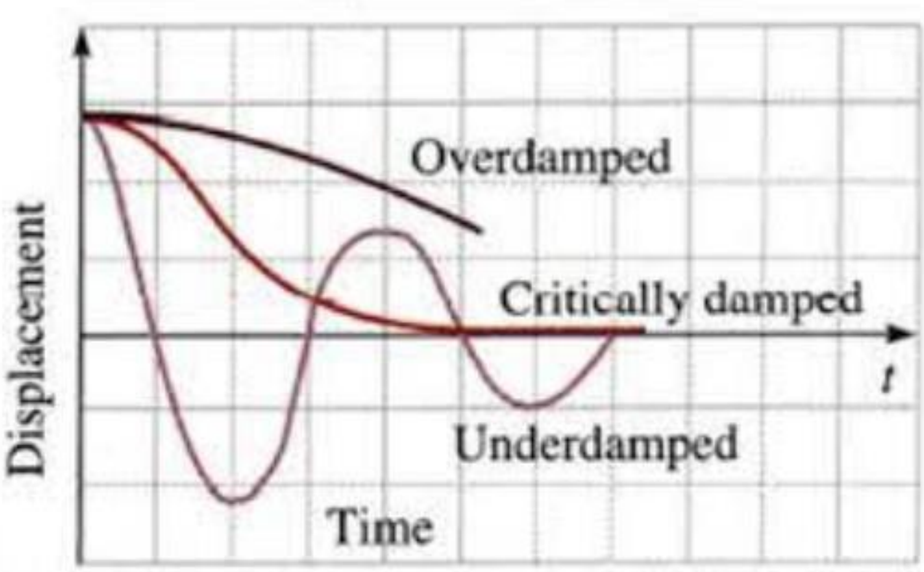
Solution of the differential equation of DHO

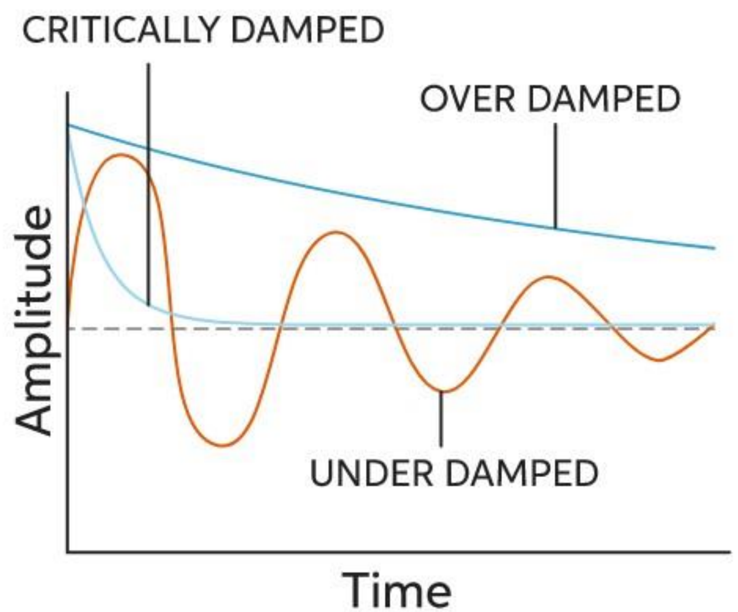
$$y = \frac{1}{2} a_0 e^{-\lambda t} [(1 + \frac{\lambda}{\sqrt{\lambda^2 - w^2}}) e^{(\sqrt{\lambda^2 - w^2})t} + (1 - \frac{\lambda}{\sqrt{\lambda^2 - w^2}}) e^{(-\sqrt{\lambda^2 - w^2})t}]$$

$$\lambda^2 > w^2 \Rightarrow Overdamped$$

$$\lambda^2 = w^2 \Rightarrow Critically\ damped$$

$$\lambda^2 < w^2 \Rightarrow Under\ damped$$





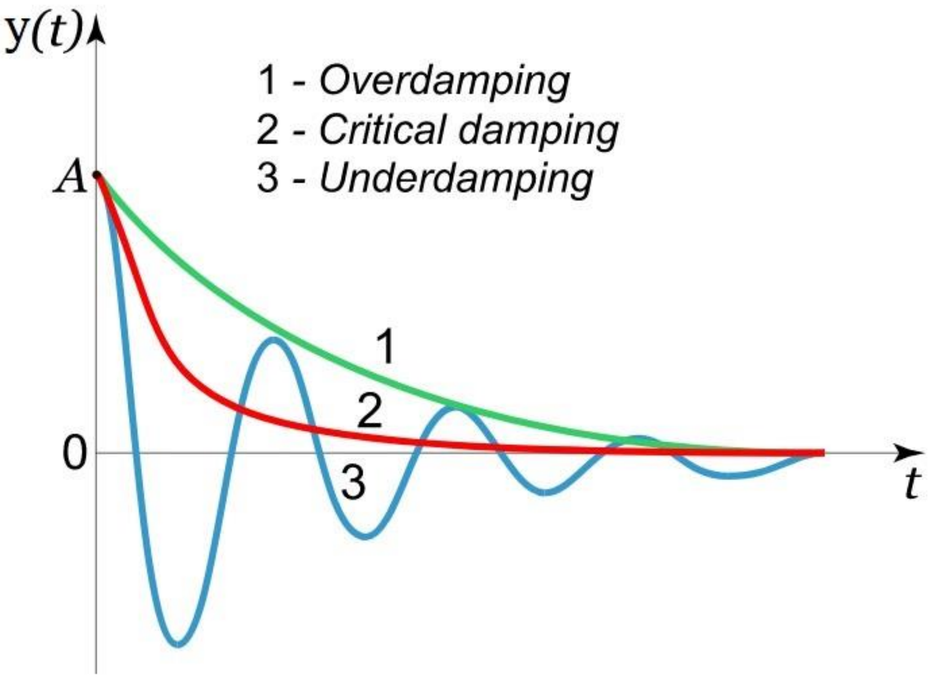
Solution of the differential equation of DHO

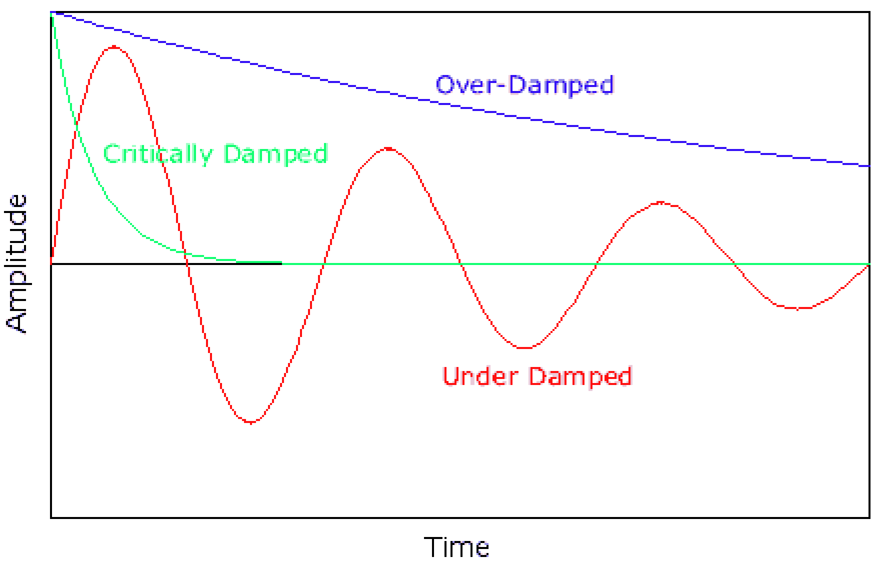
$$y = \frac{1}{2} a_0 e^{-\lambda t} [(1 + \frac{\lambda}{\sqrt{\lambda^2 - w^2}}) e^{(\sqrt{\lambda^2 - w^2})t} + (1 - \frac{\lambda}{\sqrt{\lambda^2 - w^2}}) e^{(-\sqrt{\lambda^2 - w^2})t}]$$

$$\lambda^2 > w^2 \Rightarrow Overdamped$$

$$\lambda^2 = w^2 \Rightarrow Critically\ damped$$

$$\lambda^2 < w^2 \Rightarrow Under\ damped$$





LCR Circuit

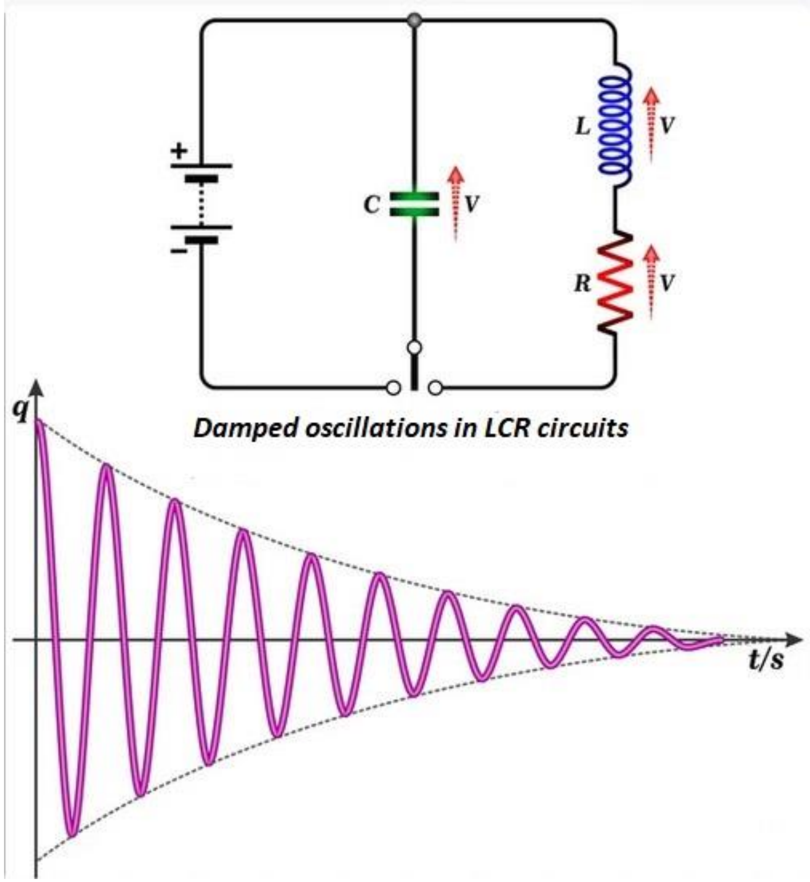
Example of Damped Harmonic Oscillation

The resistance by itself plays the part of a resistive or dissipative or damping force

$$L\frac{di}{dt}+Ri+\frac{Q}{C}=0$$

$$L\frac{d^2Q}{dt^2}+R\frac{dQ}{dt}+\frac{Q}{C}=0 \quad [\because i=\frac{dQ}{dt}]$$

$$\frac{d^2Q}{dt^2}+\frac{R}{L}\frac{dQ}{dt}+\frac{Q}{LC}=0 \quad \ldots\ldots\ldots (1)$$



Eqn. (1) is very similar to the differential eqn. of damped harmonic oscillation-

$$\frac{d^2y}{dt^2}+2\lambda\frac{dy}{dt}+\omega^2y=0$$

$$where,\ \ y\leftrightarrow Q,\ \ 2\lambda\leftrightarrow\frac{R}{L},\ \ w^2\leftrightarrow\frac{1}{LC}$$

Ref.: Physics for Engineers (Part-1) -Dr. Gias Uddin Ahmed; pp: 94-98

LCR Circuit

Example of Damped Harmonic Oscillation

The resistance by itself plays the part of a resistive or dissipative or damping force

- The RLC circuit is analogous to the damped harmonic oscillator.
- The equation of motion for the damped harmonic oscillator is:

$$m \cdot \frac{d^2x}{dt^2} + b \cdot \frac{dx}{dt} + k \cdot x = 0$$

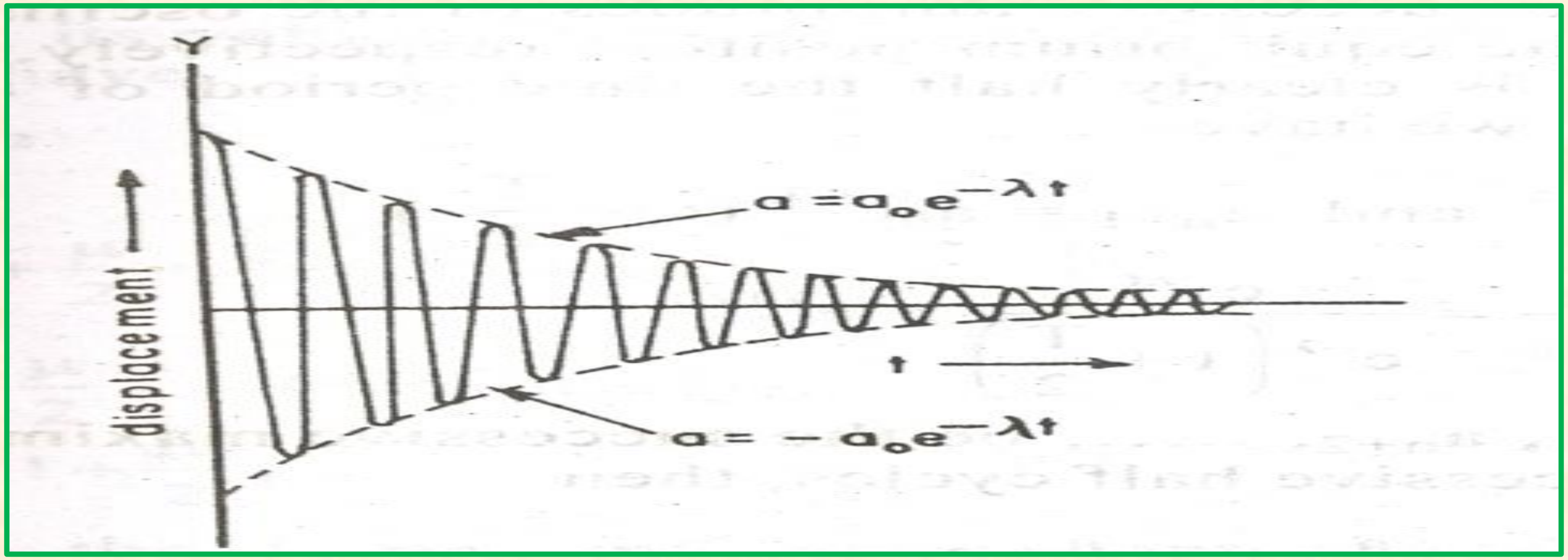
- Comparing the two equations:

$$L \cdot \frac{d^2Q}{dt^2} + R \cdot \frac{dQ}{dt} + \frac{Q}{C} = 0$$

  - Q corresponds to x; L corresponds to m; R corresponds to the damping constant b; and 1/C corresponds to 1/k, where k is the force constant of the spring.
- The quantitative solution for the quadratic equation involves more knowledge of differential equations than we possess, so we will stick with the qualitative description of the circuit behavior.

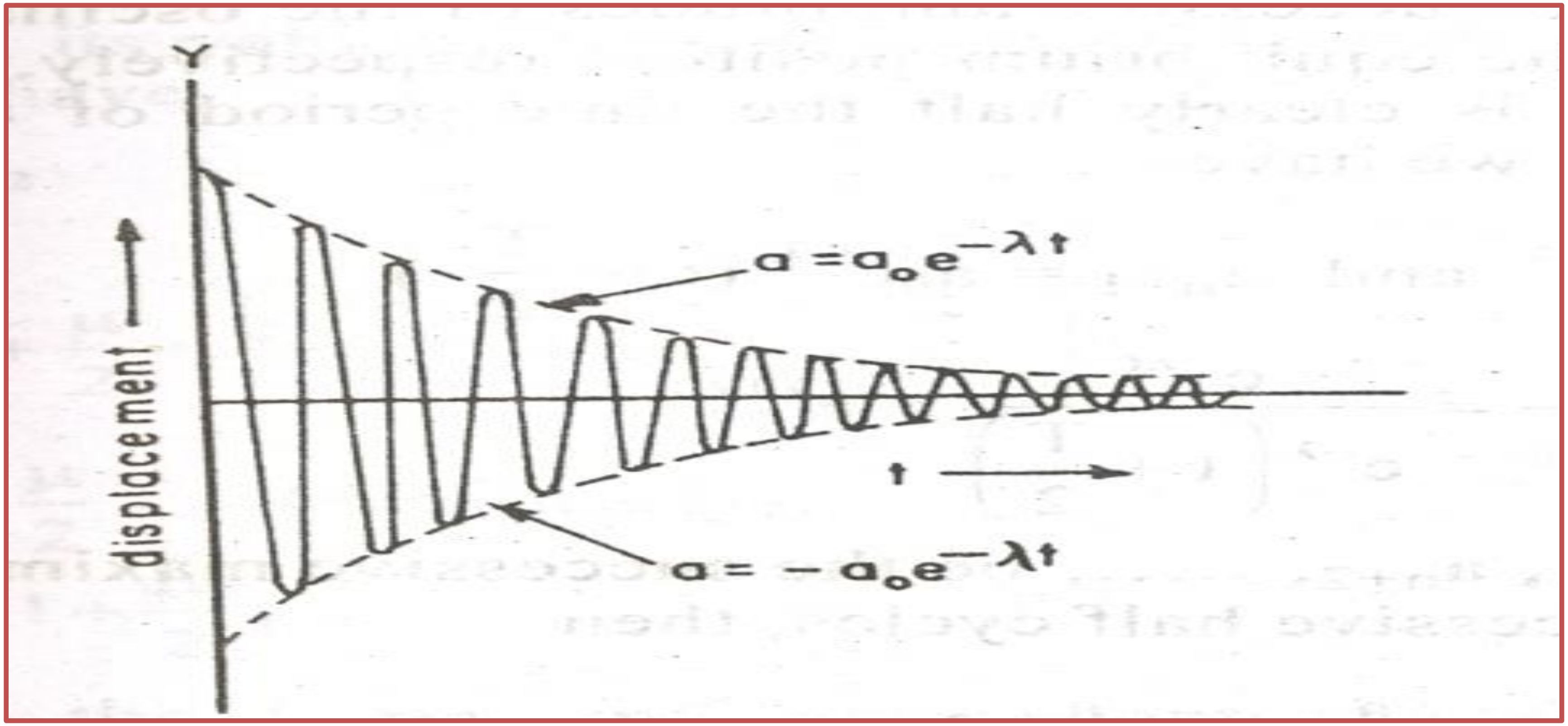
Damping produces two effects

- **The frequency of the damped harmonic oscillator is smaller then its natural or undamped frequency.**
- **The damping decreases the frequency or increase the time period of the oscillator.**



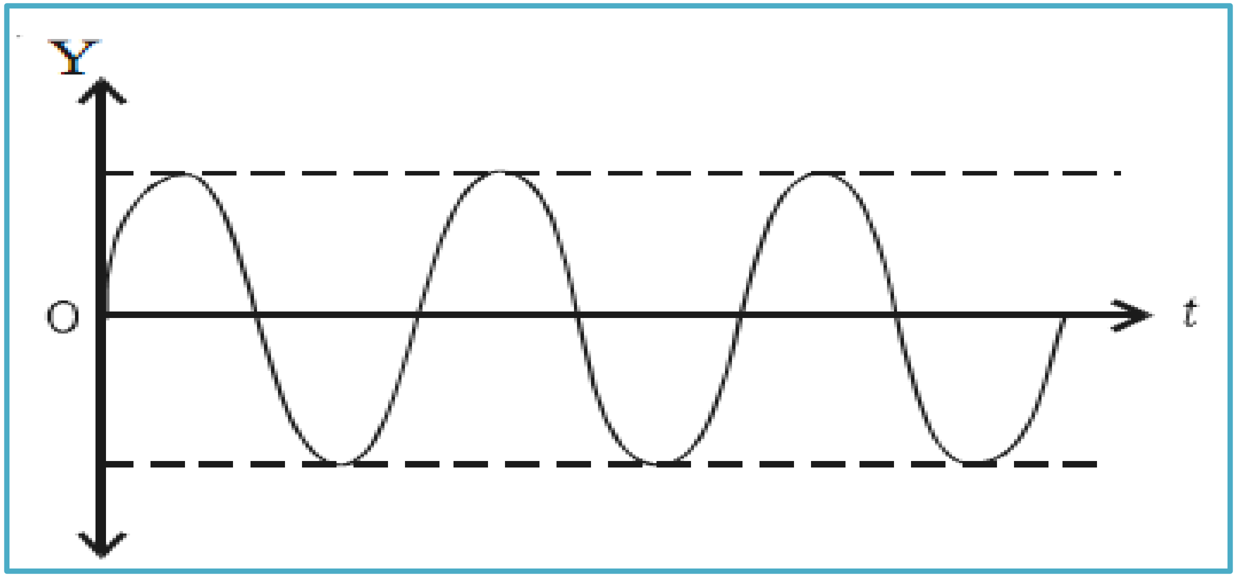
Damped oscillations

The amplitude of the oscillation does not remain constant but decay exponentially with time to zero



Maintained oscillations

- **Amplitude of an oscillating system can be made constant by feeding some energy to the system.**
- **Energy is fed to the system to compensate the energy it has lost.**



Such oscillations are called maintained Oscillations.

Types of oscillations

Forced oscillations

- The external force is driver and body is driven
- The body is forced to vibrate with an external periodic force.

Examples :

- **Sound boards of stringed instruments execute forced vibration**
- **Press the stem of vibrating tuning fork, against table. The table suffers forced vibration.**

Thank You

Thanks for your attention